

Good-Citizen Lottery

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A broader research question

How does random, as opposed to meritocratic, allocation of the awards and penalties via a lottery affect efficiency of resource allocations and social welfare?

Good-Citizen Lottery: Question

- Public bads incur social costs. We aim to collectively minimize the production of public bads.
- “Evil pays more”: Good citizens are typically unpaid for their good deeds.

Question: Would a lottery paying one of the **good citizens** whose prize is funded by bad citizens be effective in reducing social costs?

Motivation

- Check this [partially fake news](#).



There's a speed camera lottery in Stockholm, Sweden where drivers who drive at or under the speed limit are entered to win money. The prize fund comes from the fines paid by people who were speeding.

Greenhouse gas emissions, littering, illegal parking, and speeding are examples of situations where good citizen behaviors are costlier.

(Example that I want to avoid: Voter turnout lottery funded by penalties collected from no turnouts)

Previous Studies

It is not rare to use lotteries for nonstandard situations.

- Kim (2021): vaccination lottery
- Kim (2023): penalty lottery
- Gerardi et al. (2016), Duffy and Matros (2014): turnout lottery
- Kearny et al. (2010), Filiz-Ozbay et al. (2015): savings lottery
- Morgan (2000), Morgan and Sefton (2002): lottery to fund public goods
- Björkman Nyqvist et al. (2018): lottery incentivizing safer sexual behavior
- Volpp et al. (2008), Levitt et al. (2016): lottery for habit formation

Setup

(Can be more general)

- n citizens
- Each player chooses either S (safely abide by law) or V (violate it).
- Citizen i accrues a benefit of acting V , $B_i > 0$.
- Incomplete monitoring capacity: With probability $p \in (0, 1)$, action V is monitored and fined F . p is fixed.
- Assume $B_i - pF > 0$ for all i , so bad behavior is beneficial. (Otherwise the question becomes trivial.)
- The payoff of choosing S is 0 for 'most' cases, but when selected as a winner of the lottery, it is kF , where k is the number of players who chose V and got monitored.

Illustration

- Consider three homogeneous citizens, $B_i = B$ for all i .
- sort of a coordination game: neither (S, S, S) nor (V, V, V) is an equilibrium (under some parametric restrictions).
- Given two others playing S : I get 0 if I play S , and I get $B - pF > 0$ if I play V , so I should play V .
- Given two others playing V : I get $2pF$ if I play S , and I get $B - pF$ if I play V , so as long as $p > \frac{B}{3F}$, I should play S .
- A symmetric MSNE: playing S with probability δ^* , where

$$\delta^* = \frac{3 - \sqrt{4B/pF - 3}}{2}$$

(Note that we assume $B - pF > 0$ so $4B/pF > 4$.)

Analysis

For $n + 1$ citizens, given each plays S with prob δ ,

- the expected payoff of playing S :

$$\begin{aligned} & \binom{n}{0} \delta^n (1 - \delta)^0 0F + \binom{n}{1} \delta^{n-1} (1 - \delta) \frac{pF}{n} + \binom{n}{2} \delta^{n-2} (1 - \delta)^2 \frac{2pF}{n-1} + \cdots + \binom{n}{n} \delta^0 (1 - \delta)^n n pF \\ &= \sum_{i=1}^n \binom{n}{i} \delta^{n-i} (1 - \delta)^i \frac{i pF}{n + 1 - i} \end{aligned}$$

the expected payoff of playing V : $B - pF$

- $\delta \in (0, 1)$ such that

$$B - pF = \sum_{i=1}^n \binom{n}{i} \delta^{n-i} (1 - \delta)^i \frac{i pF}{n + 1 - i}$$

is a MSNE strategy.

- Let $b := \frac{B}{pF} - 1 \in (0, 1)$, the normalized excess benefit to the expected cost.

$$b = \underbrace{\sum_{i=1}^n \binom{n}{i} \delta^{n-i} (1 - \delta)^i \frac{i}{n + 1 - i}}_{R(\delta, n)}$$

Analysis

Lemma: $R(\delta, n) = \sum_{k=1}^n (1 - \delta)^k$. $R(0, n) = n$, $R(1, n) = 0$, and $R(\delta, n)$ monotone decreases in δ .

Proof sketch: $R(\delta, n) = \sum_{i=1}^n \binom{n}{i} \delta^{n-i} (1 - \delta)^i \frac{i}{n+1-i}$. Replace $\binom{n}{i} \frac{i}{n+1-i} = \frac{i}{n+1} \binom{n+1}{i}$. Apply the mean of *Binomial*($n+1, 1 - \delta$) is $(n+1)(1 - \delta)$. Apply $\sum_{k=0}^{n-1} (1 - \delta)^k = \frac{1 - (1 - \delta)^n}{1 - (1 - \delta)}$.

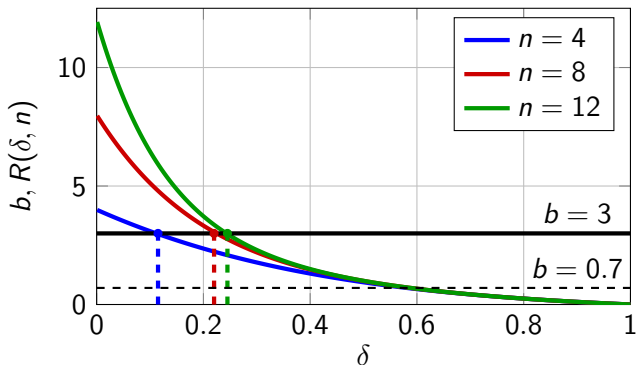
The equilibrium probability of good-citizen behavior δ is such that

$$b = \sum_{k=1}^n (1 - \delta)^k, \quad (1)$$

which is analytically tractable.

Analysis

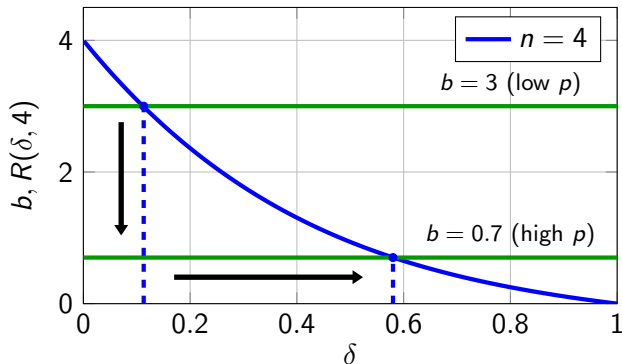
Proposition 1: Given $b \in (0, n)$ and $n \in \mathbb{N}_+$, let $\delta^*(n)$ be the solution to $b = \sum_{k=1}^n (1 - \delta)^k$. $\delta^*(n)$ is unique and increasing in n .



- Intuition: Larger n decreases the winning probability and increases the expected payoffs of winning. Proposition 1 shows that the upside force dominates the downside one.
- Prop 2: When $b \in (0, 1)$, $\delta^*(n)$ stabilizes exponentially fast.

Analysis

Proposition 3: $\frac{d\delta^*}{dp} > 0$ for $b \in (0, n)$.



- Larger $p \Rightarrow$ more likely to play S , as long as $b \in (0, n)$.
- Intuition: The excess benefit decreases. Very straightforward.

Analysis

There are some factors not yet analyzed but worth mentioning.

Risk preferences

- ⇒ It is not straightforward to tell which action is more “risky”.
For sufficiently large n , risk-averse subjects might play S more.
The winning prob. of the lottery is negligible, while the payoffs of V are volatile. For small n , V may be more attractive.

Subjective probability weighting

- ⇒ Some people subjectively overestimate small probabilities.
Those who **overestimate the small probability of winning the lottery would be more willing to play S .**

Testable Hypotheses

$\delta(n)$: The fraction of subjects playing S when the group size is n .

- ① (Regarding Proposition 1 and 2) When b is large, $\delta(n)$ is increasing in n . When b is small, $\delta(n)$ is relatively stable in n .
- ② (Regarding Proposition 3) $\delta(n)$ is greater with a higher p .
- ③ Risk aversion has an ambiguous effect on $\delta(n)$.
- ④ The subjects with a stronger tendency of overestimating small probabilities would prefer to play S .

Experimental Design

Case of 18 subjects, $p = 0.3$

Varying n within subjects, varying p between subjects. In each session of 18 subjects, they play 12 similar games, wherein:

- A subject is randomly assigned to a group whose size is $n \in \{3, 6, 9, 18\}$. Their task is to choose one of the two items: a white ball (corresponding to S) and a box (V).
- Unwrapping the box, a subject gets a red ball with probability $p = 0.3$ and a blue ball with probability $1 - p$.
- By getting a blue ball, a subject earns a payoff of 240. A red ball is associated with a payoff of 40.
- Choosing the white ball earns 100. Also, one of the group members who chose the white ball is randomly selected to get an additional payoff of $200k$, where k is the number of the members who got the red ball.
- The experiment currency unit is tokens. (1 token = 100 KRW)

Experimental Design, cont'd

The subjects repeat the game in the mixed order in terms of n .

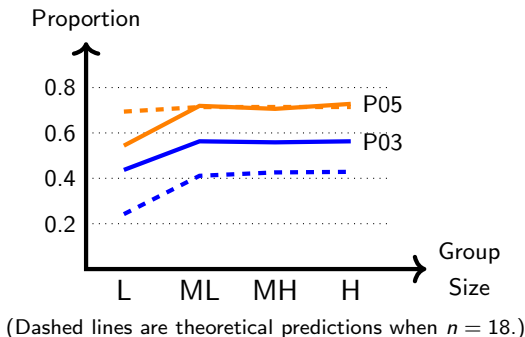
Round	1	2	3	4	5	6	7	8	9	10	11	12
n (when $N = 18$)	6	9	18	3	9	6	18	3	9	6	3	18
n (when $N = 20$)	5	10	20	2	10	5	20	2	10	5	2	20
Choosing a box $\Rightarrow$ 40 with prob p ; 240 with prob $1 - p$. Choosing a white ball $\Rightarrow$ Randomly selected one additionally earns 200k.												
$p = 0.3$ or $p = 0.5$												

- Given those parameters, $b = \frac{B-pF}{pF} = \frac{4}{3} > 1$ when $p = 0.3$, and $b = \frac{2}{5} < 1$ when $p = 0.5$.
- Theory predicts that $\frac{\Delta\delta(n)}{\Delta n} > 0$ in P03 (high b) treatment, while $\frac{\Delta\delta(n)}{\Delta n} \approx 0$ in P05 (low b) treatment.
- Then, risk preference and subjective probability weighting are elicited using a simple survey. Post-experimental survey include typical things as well.
- One of the 12 rounds is randomly selected to be paid.

Experiment Procedure

- Zoom-administered real-time online experiment
- LIONESS (Live Interactive ONline Experimental Server Software)
- at SKKU, in Korean
- 4 sessions each for P03 and P05. $74 + 76 = 150$ participants
- Random regrouping
- On average 18,800KRW; min 5,000KRW, max 75,000KRW.
- Starbucks e-gift cards corresponding to the cash value

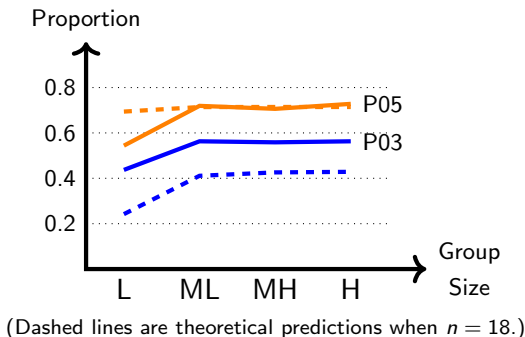
Result 1



- In P03, the proportion of ball choices is significantly larger than theoretical prediction ($p = 0.003$). The proportion increasing in a statistically nonsignificant manner ($p = 0.148$).
- In P05, the observed proportion is statistically indifferent from the theoretical prediction ($p = 0.1091$). The proportion increases in the group size ($p = 0.031$).

These two partially support Hypothesis 1.

Result 2

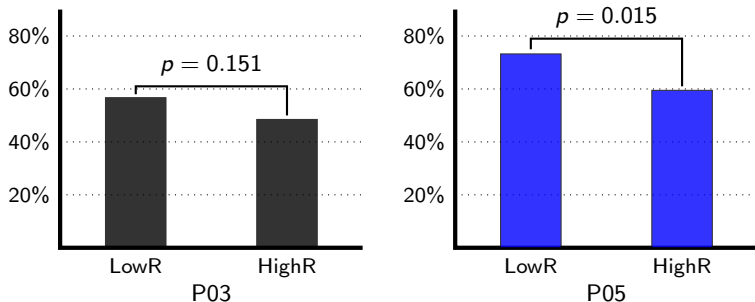


- The proportion of ball choices in P05 is significantly larger than that in P03 ($p < 0.001$), supporting Hypothesis 2.

What/who drives S then, especially in P03?

Result 3

Proportion of Ball Choices By Risk Preference

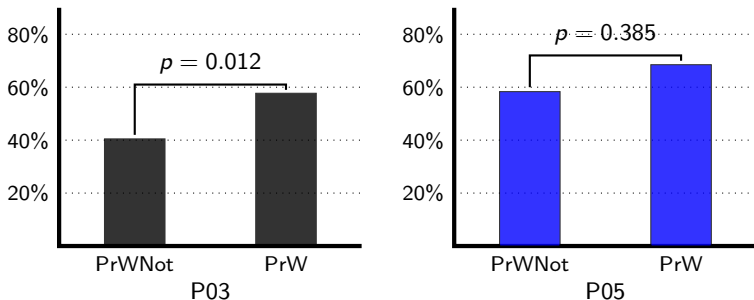


* HighR: high risk-taking tendency.

- More risk averse $\Rightarrow$ more likely to choose S, though the difference is statistically insignificant in P03.

Result 4

Proportion of Ball Choices By Probability Weighting Tendency



* PrW: exhibiting a tendency of probability weighting.

- Those who tend to subjectively weight small probabilities more are more likely to choose S , supporting Hypothesis 4.
- ⇒ Citizen lottery may work in a large population with a low enforcement (low p) as well.

Take-Away Messages

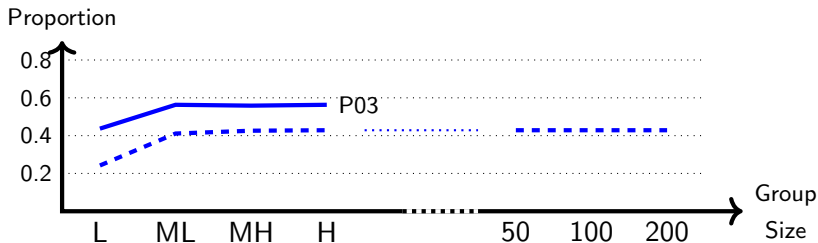
- Theory predicts (and experimental findings show) that the proportion of good citizens
 - increases/stabilizes (weakly increases) in the group size;
 - increases (increases) in monitoring capacity;
 - is ambiguous (weakly increases) in risk aversion;
 - increases (increases) in probability-weighting tendency.
- Experimental findings are largely consistent with theoretical predictions.
- Result 4 is good: It suggest that the citizen lottery may work even for a large population, without increasing the monitoring capacity.

New supplementary evidence

A common criticism: A group of (at most) 20 doesn't seem like a "large" group. Would we observe the similar results when a group size becomes really large?

- I conducted an online experiment at Prolific. The data from 400 subjects were collected on Feb, 2025.
- The same setup with P03, but the group sizes are 50, 100, and 200. (Orders were randomized.)

Proportion of playing S: Laboratory vs. Online Extension



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